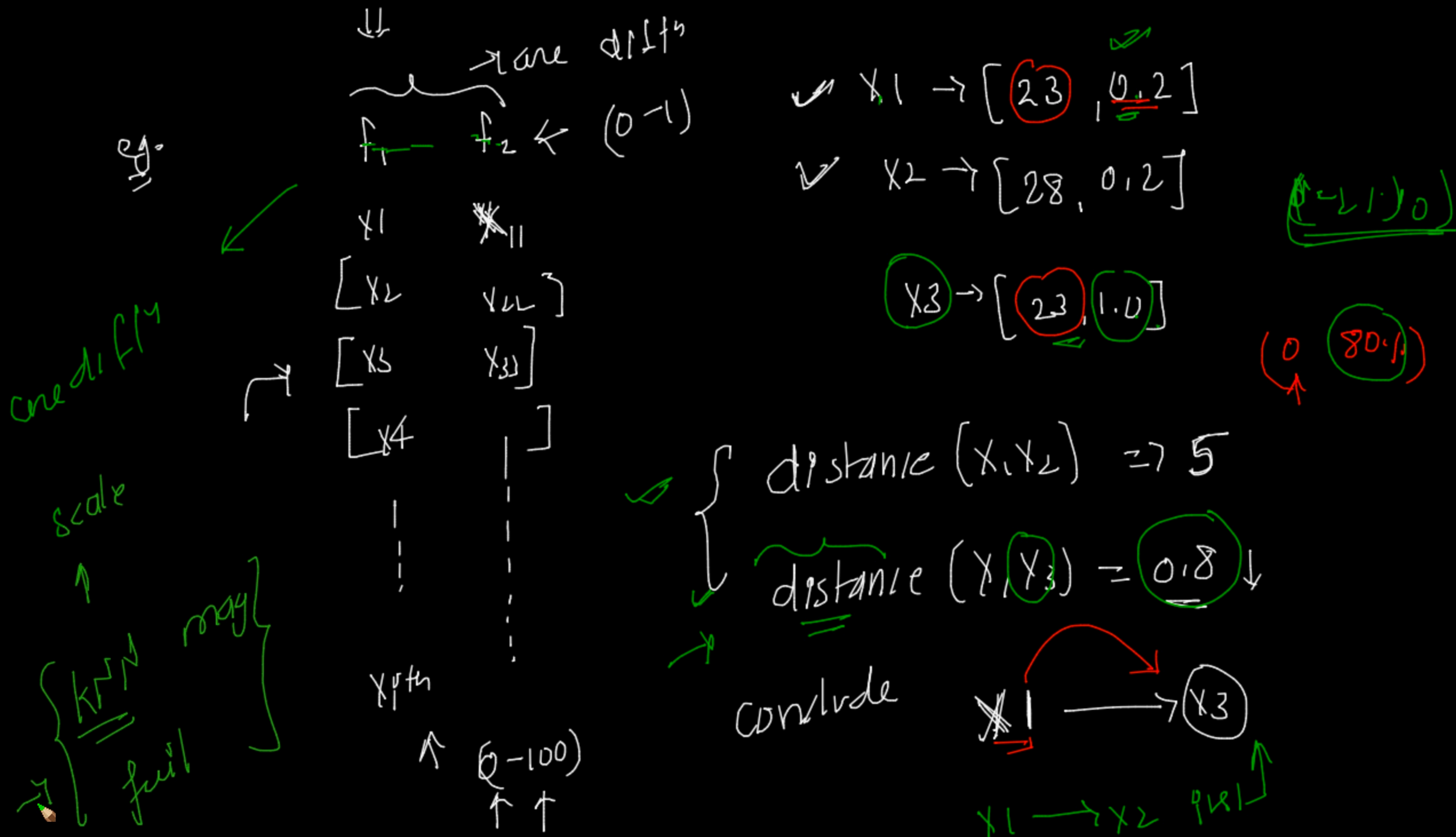


Importance of scale and column std.



Before applying KNN
↑↑↑

Data Standardization → (-1)

$$\text{mean} = 150$$

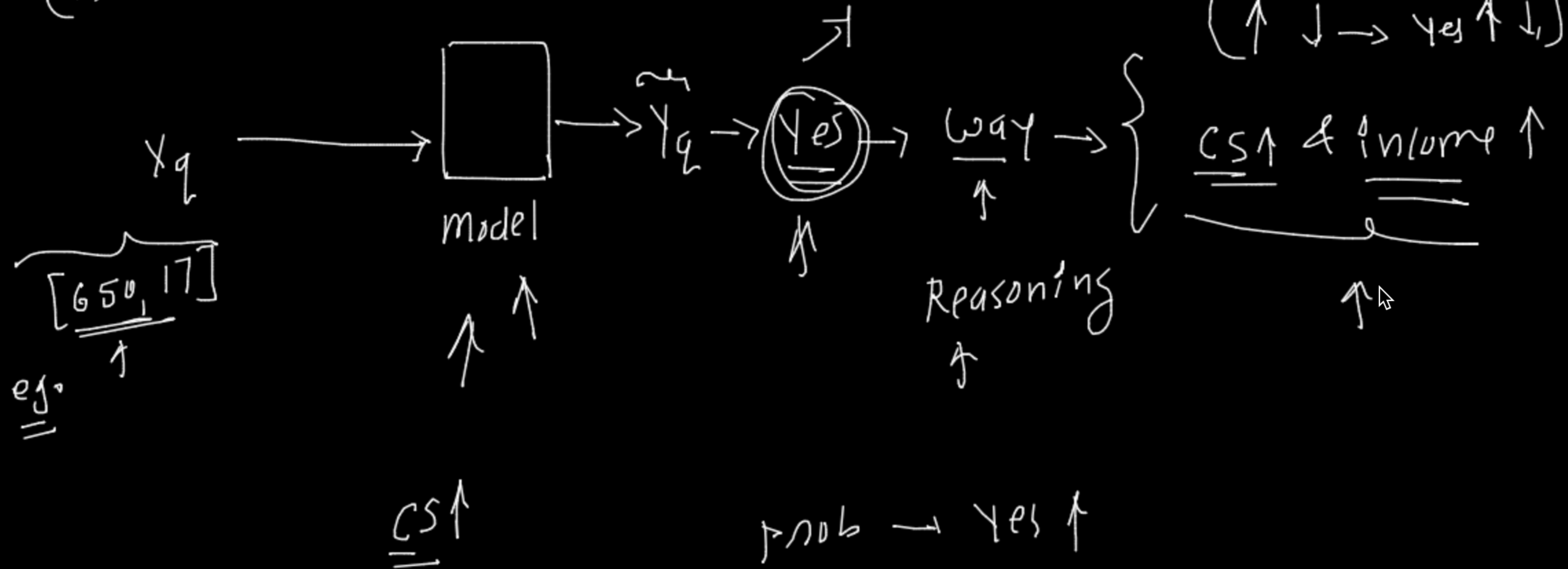
$$\text{std} = \bar{x}$$

→ col → $\text{col}' \rightarrow (0-1)$

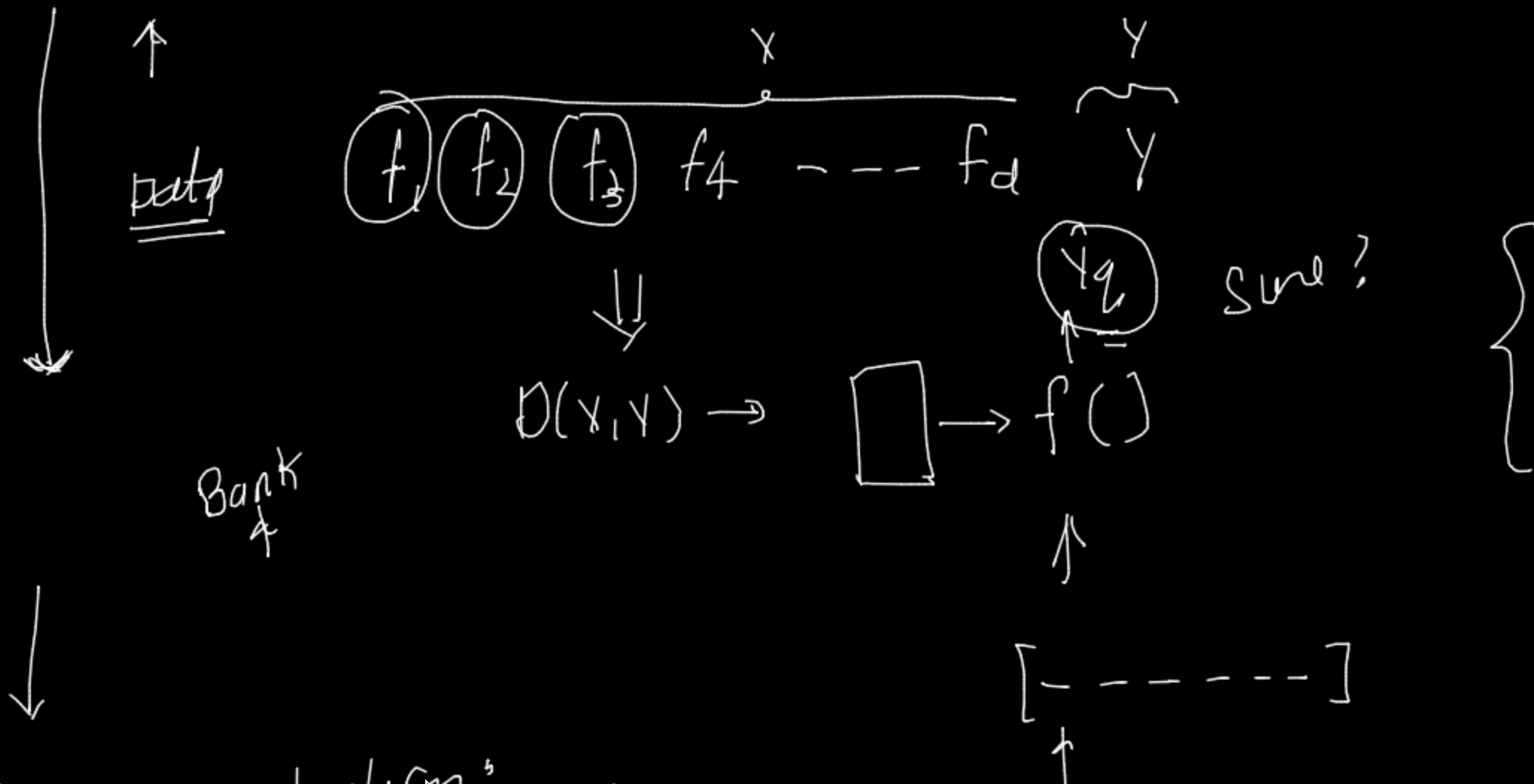
$$\begin{array}{l} 100 \rightarrow (100 - 150) / \text{std} \\ 110 \\ 120 \\ 156 \\ 175 \\ \vdots \\ \vdots \\ \vdots \end{array}$$

Model Interpretability vs 'BlackBox'

(i)



Feature Importance



'Feature selection'

$L_1(i) \underline{RF} \vec{E} (KNN)$

(i) KNN → Feature importance →  → 'BlackBox' }

$$D_n = \{T_H, T_e\}$$

F	y



T _H	T _e
f ₁	



f₁ → a₁



T _H	
f ₂	



f₂ → a₂

f₄

T _H	T _e
f ₃	f ₁



a₁

f ₃	f ₂



f₁

↓
a₁

f₂

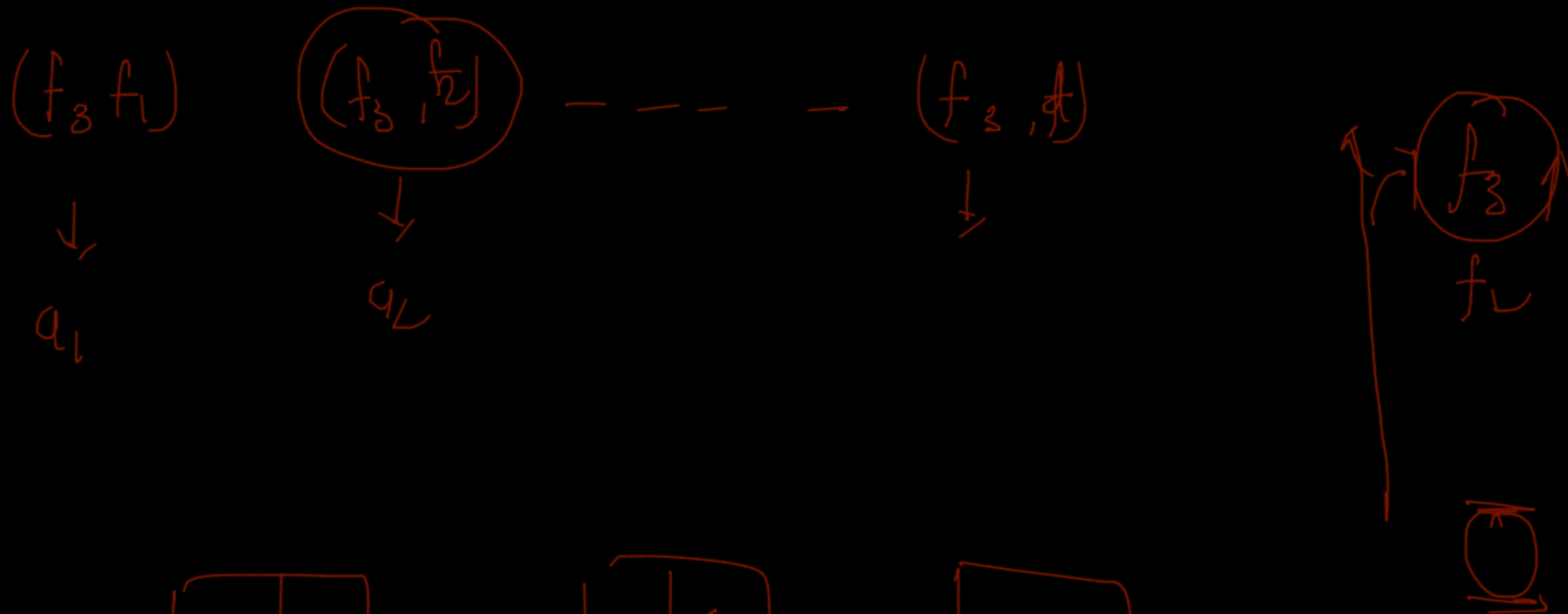
↓
a₂

f₃
↓
a₃

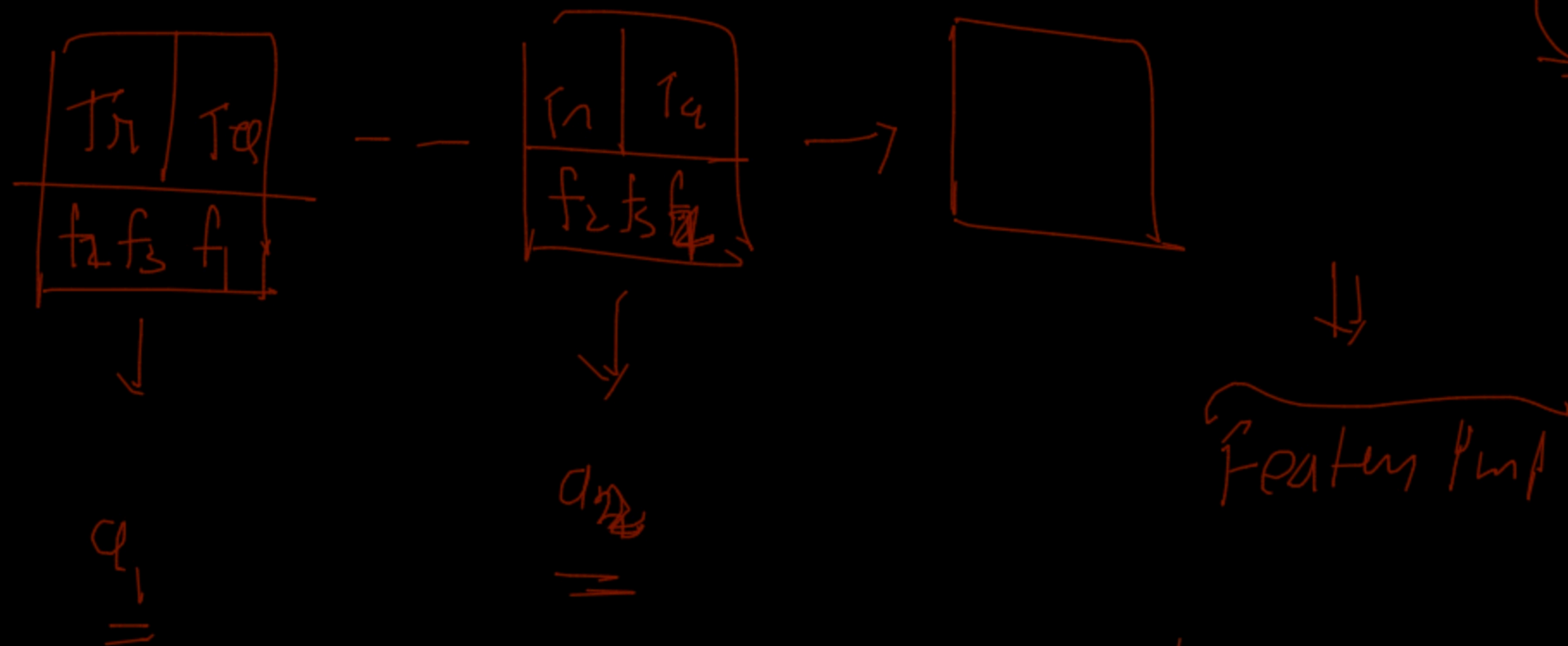
78%

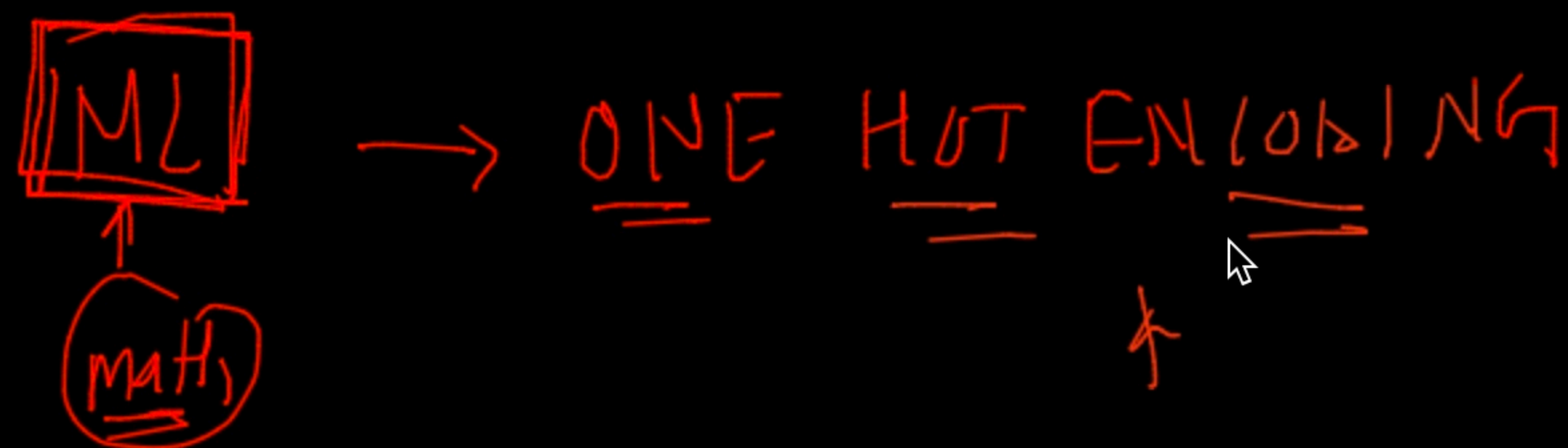
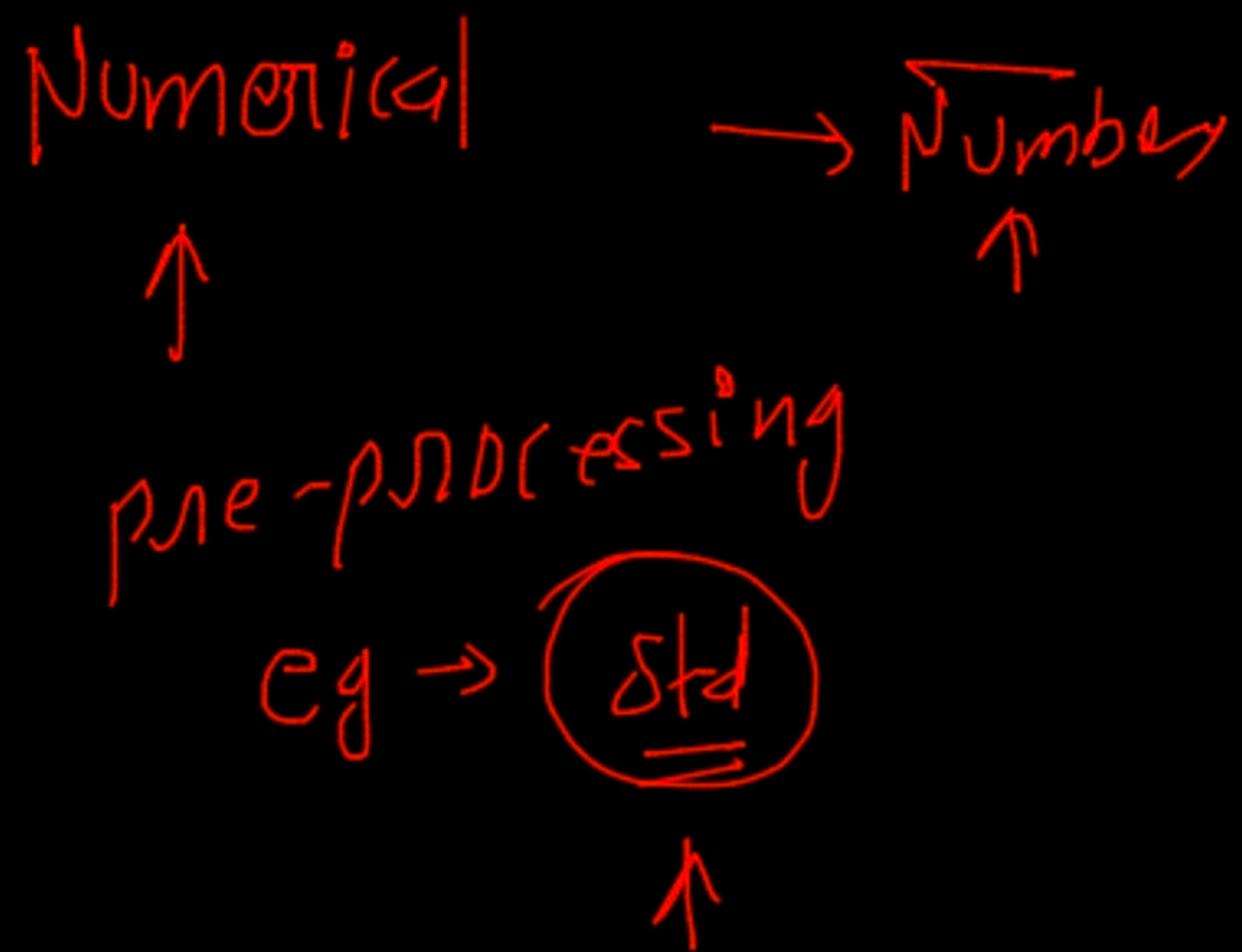
f_d

↓
a_d



step 3

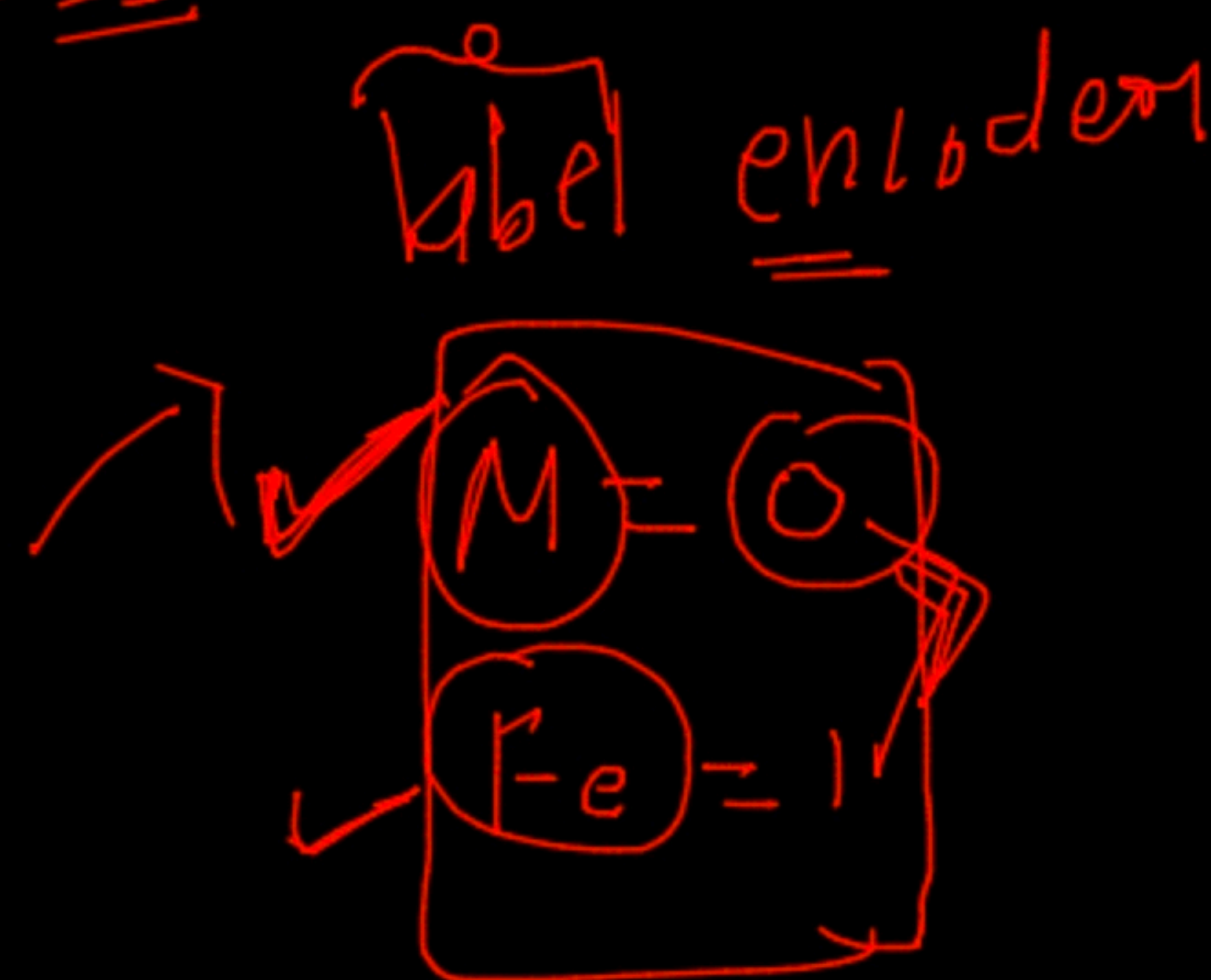
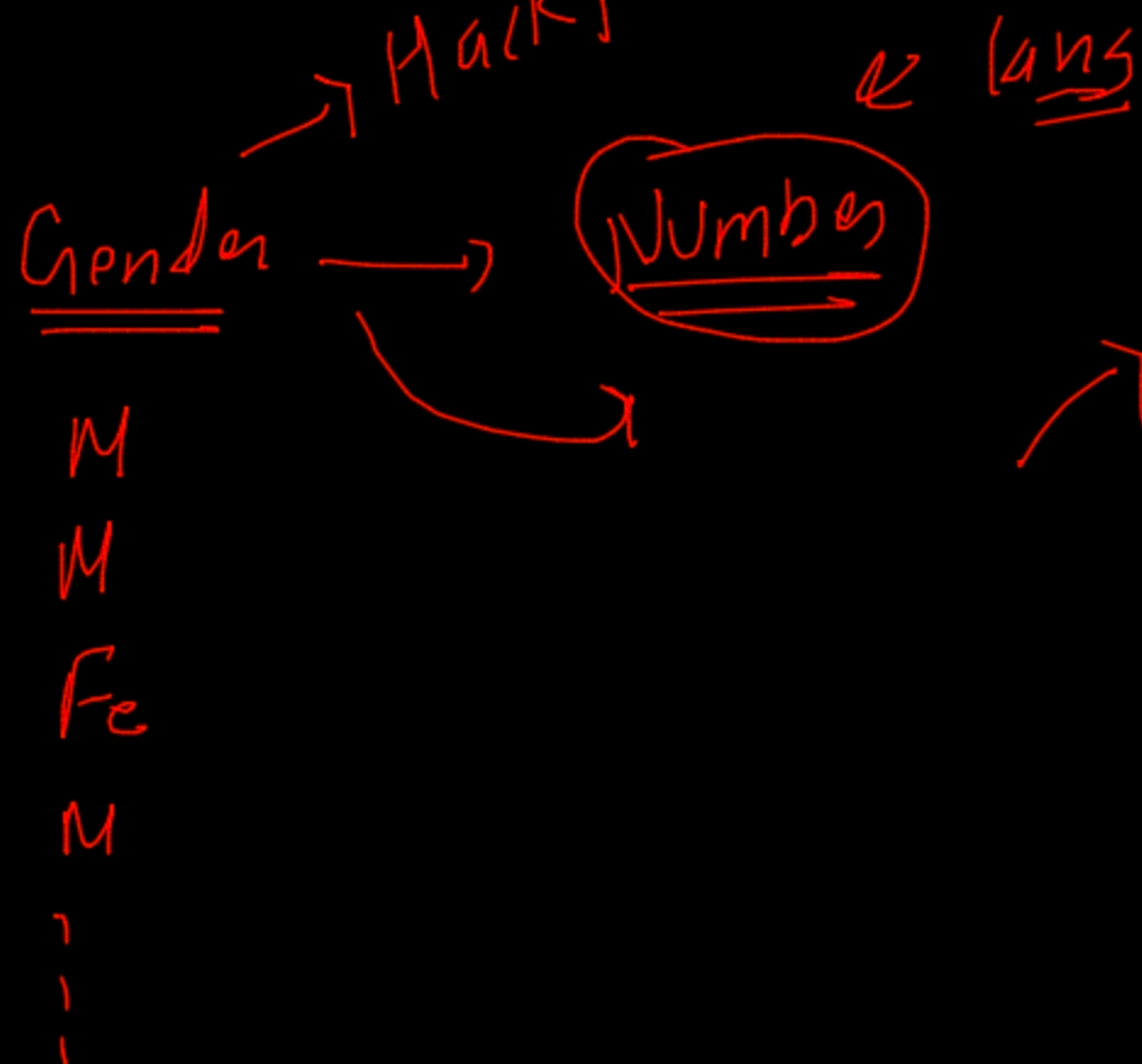




Sklearn $\rightarrow$ Hack/technique

Assume

Encode



$0 < 1$

$1 > 0$

$\vdots$

Gender



M

F

=



Gender

M

M

F

F

M	F
1	0

DNE not encoding

M	F
1	0
1	0
0	1
0	1

UNK^W



✓	✓	✓	✓

City

Tier-I

Tier-II

⋮

Tier-III

ONEHOT
ENCODING
→

↓

(d↑)
=

Tier-I

Tier-II

Tier-III

1

0

0

0

1

0

1

⋮

1

1

1

1

Country

|||||

(2nd)

Blackboard

(↑)